

# Assignment

## Module 1.4 GPU and Cooling system

1. Open your own laptop or desktop (or use an online PC hardware simulator if you don't have access to a real system) and identify the GPU model installed. Write down the GPU's brand, model number, and whether it is integrated or dedicated.



**GPU Brand:** Intel

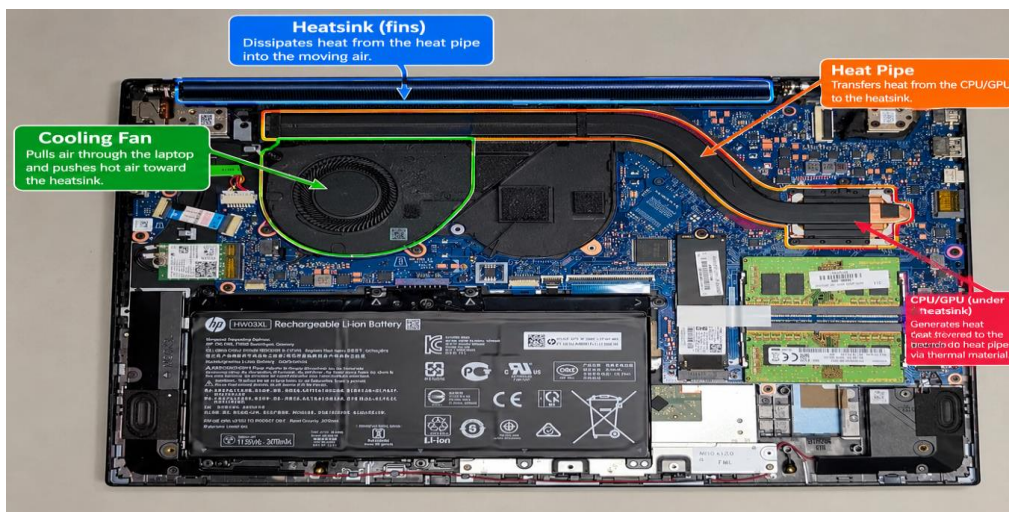
**GPU Model:** Intel Iris Xe Graphics

**GPU Type:** Integrated GPU

My laptop has Intel Iris Xe Graphics, made by Intel. This is an integrated GPU, which means it is built directly into my laptop processor instead of being a separate graphics card. It shares my computer's regular memory for graphics tasks rather than having its own dedicated memory.

Intel Iris Xe Graphics works well for my everyday tasks like web browsing, watching videos, joining online classes, using Microsoft Office, and casual gaming. It can also handle some photo and video editing, although it is not as powerful as a dedicated GPU for demanding games and graphics work. Integrated graphics also have some benefits because they use less power and allow my laptop to be thinner and more battery efficient.

2. Take a clear photo or screenshot of the cooling system (fan, heatsink, or liquid cooling setup) inside your PC or from a PC hardware simulator. Label the main cooling components and briefly describe their function.



My laptop has an air-cooling system that keeps the CPU and GPU from getting too hot. This system has three main parts: a cooling fan, a copper heat pipe, and a heat sink.

**Cooling Fan:** The fan draws air into the laptop and pushes warm air out. This moving air helps take heat away from the cooling system.

**Copper Heat Pipe:** The heat pipe moves heat away from the CPU and GPU. It carries this heat to the heatsink, where the heat can escape.

**Heatsink:** The heatsink has many thin metal fins that create a large area for cooling. Air from the fan flows over these fins and removes the heat.

**Thermal Paste/Pad:** A special material sits between the processor and heatsink to help heat move from the CPU/GPU to the cooling system.

**3. Create a comparison table listing at least 3 differences between air cooling and liquid cooling systems for GPUs, including factors like cost, efficiency, and maintenance.**



| Factor       | Air Cooling                                                                          | Liquid Cooling                                                                                           |
|--------------|--------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| Cost         | Air cooling is usually cheaper and does not require many extra parts.                | Liquid cooling is more expensive because it needs a pump, radiator, and coolant.                         |
| Efficiency   | Air cooling provides good cooling for normal gaming and everyday use.                | Liquid cooling can cool the GPU more effectively, especially when it is under heavy load.                |
| Maintenance  | It is easy to maintain. We mainly need to clean the fans and remove dust.            | It needs more maintenance because we have to check the pump, radiator, and coolant.                      |
| Noise        | The fans can become loud when the GPU gets hot.                                      | It can be quieter in some systems, but the pump and radiator fans can still make some noise.             |
| Installation | It is easier to install because the cooler is simple and usually comes with the GPU. | It is harder to install because we need to connect several parts such as the pump, radiator, and tubing. |
| Risk         | It has a low risk because there is no liquid that can leak.                          | There is a small risk of leakage or pump failure if the system has a problem.                            |

**4. Imagine you are building a gaming PC for streaming IPL matches and playing graphics-heavy games. Choose between air cooling and liquid cooling for your GPU and justify your choice in 3-4 sentences based on your research.**



I would pick liquid cooling for my GPU since I want to stream IPL matches and play high-graphics games on my gaming PC. When we run demanding games or streams for extended periods, our GPU generates significant heat, so effective cooling matters. Liquid cooling removes heat quickly and maintains steady GPU temperatures during intense use. It can also lower fan noise, though it costs more and requires careful setup and upkeep. In the end, I'd

choose liquid cooling because it delivers excellent cooling power for my gaming and streaming activities.